第十章
气体的压缩
Compression of Gas



作业

教材习题：

10-3、10-4、10-6

补充习题：

11-11

第十章 气体的压缩

§ 10-1 压气机工作原理

§ 10-2 压缩过程的热力学分析

§ 10-3 活塞式压气机余隙容积的影响

§ 10-4 多级压缩及中间冷却

压气机的作用和分类

压气机的作用

提高发动机进气压力增加进气量

生活中：自行车打气

工业上：锅炉鼓风、出口引风

型式结构

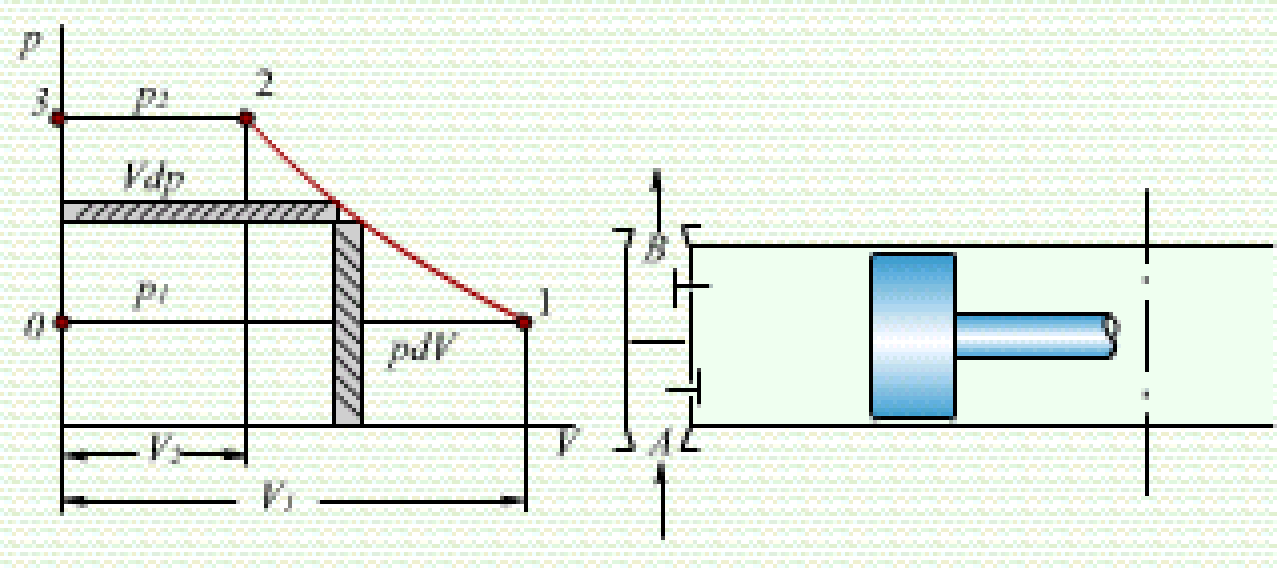
- 活塞式（往复式）
- 叶轮式
  - 离心式
  - 轴流式

压力范围

通风机	$\Delta p < 0.01MP_a$	**Fan**
鼓风机	$0.01MP_a < \Delta p < 0.3MP_a$	**Blower**
压缩机	$\Delta p > 0.3MP_a$	**Compressor**

§10-1 压气机工作原理

活塞式压气机



0 - 1： 吸气过程

1 - 2： 压缩过程 ⟶ 热力过程 $\dfrac{p_2}{p_1}=\pi$ 增压比

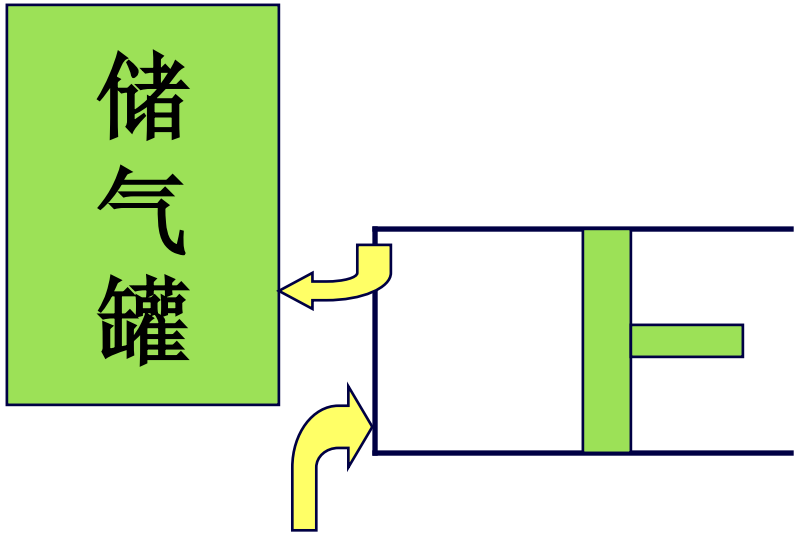
2 - 3： 排气过程

活塞式压气机的特点

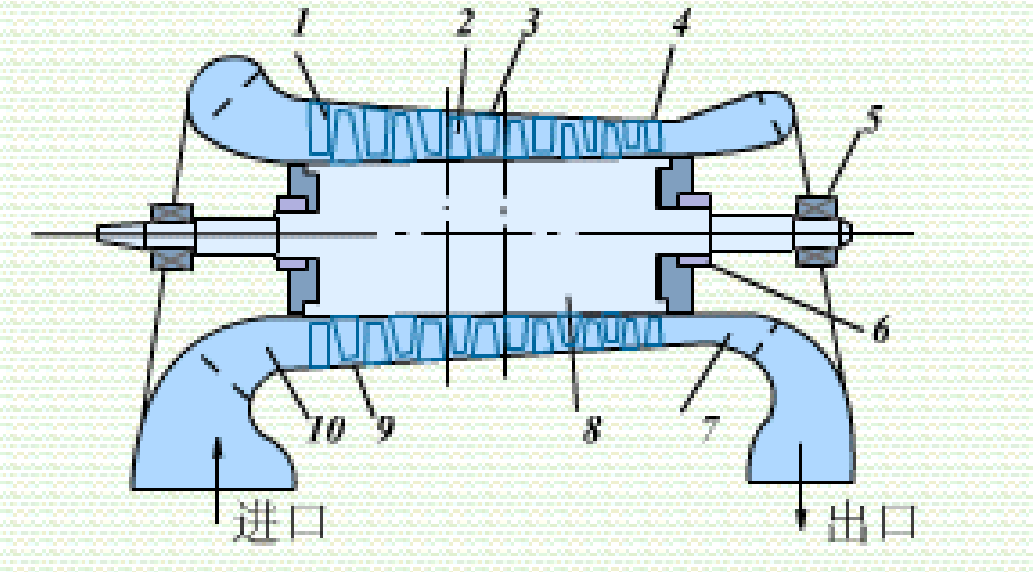
流量小，升压幅度可以很大

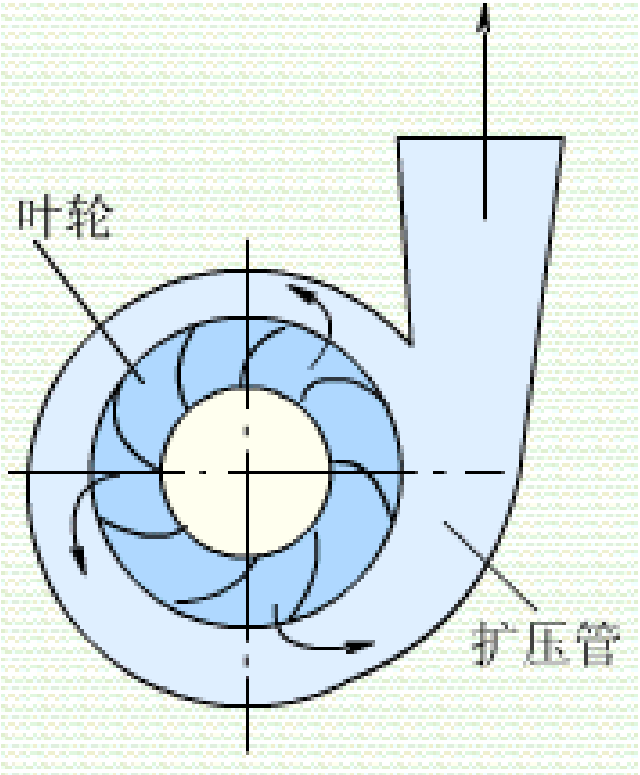
普遍应用于汽车、电冰箱及化工领域





叶轮式压气机





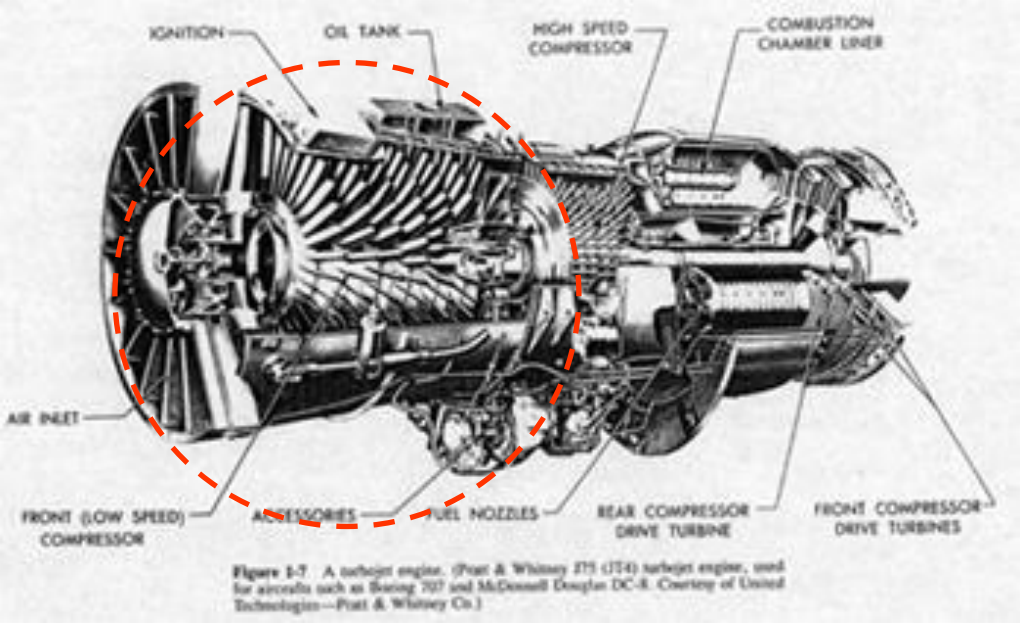
轴流式压气机

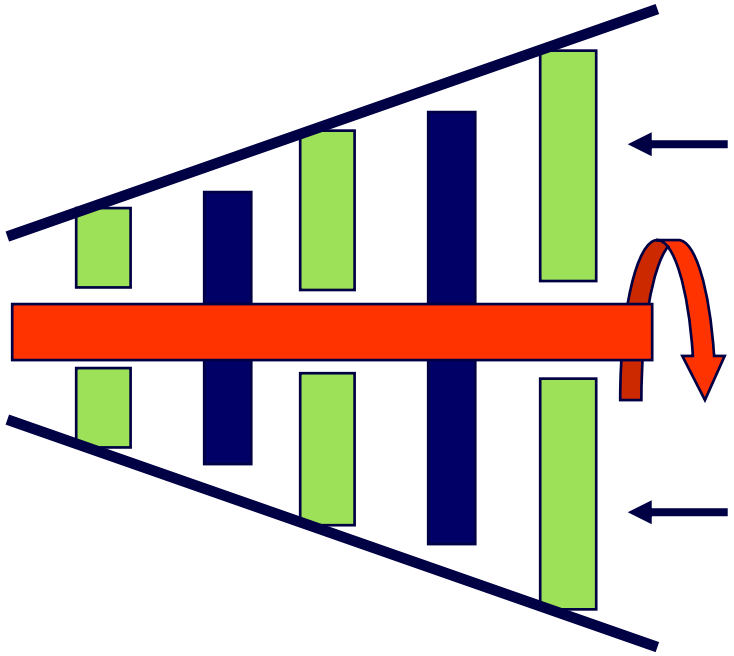
离心式压气机

轴流式压气机的特点

流量大，结构复杂

普遍应用于飞机、轮船及化工领域

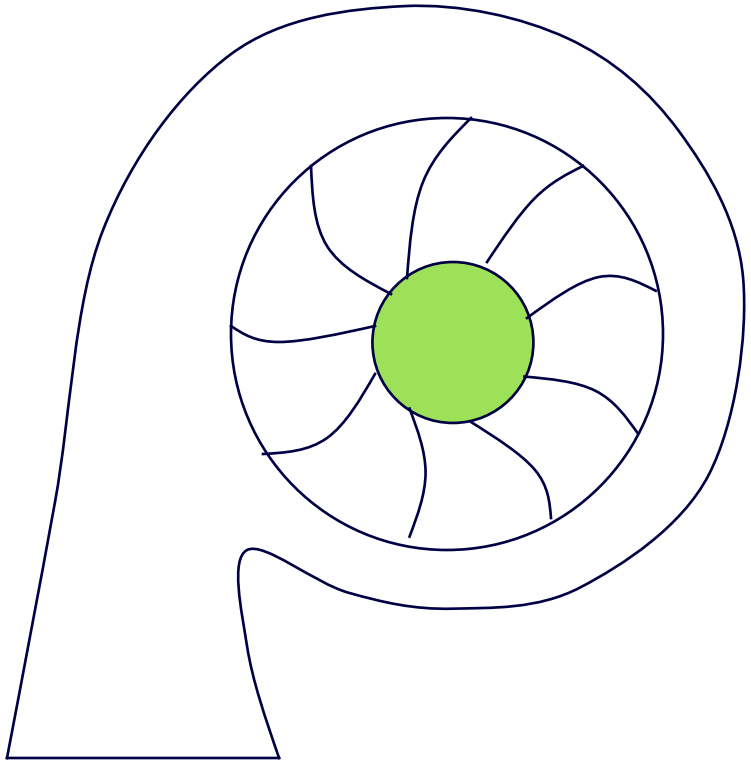




大型离心式风机及其特点

升压幅度小，结构简单





活塞式与叶轮式压气机的异同

相同：状态变化过程

不同：结构、工作原理

活塞式压气机：

间歇性，低转速，小排量、高压力

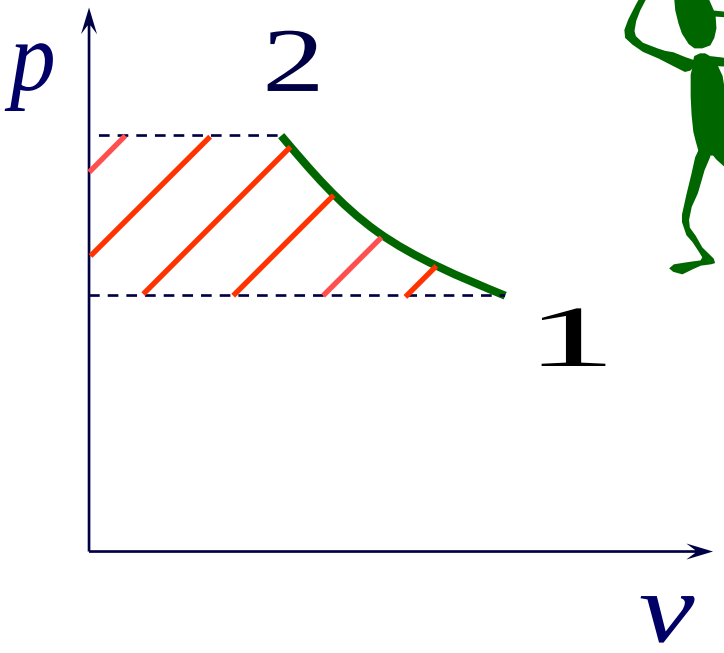
叶轮式压气机：

连续性，高转速，大排量、低压力

要想提高压力，必须多级压缩

§10-2 压缩过程的热力学分析

目的：研究耗功，越少越好





指什么功

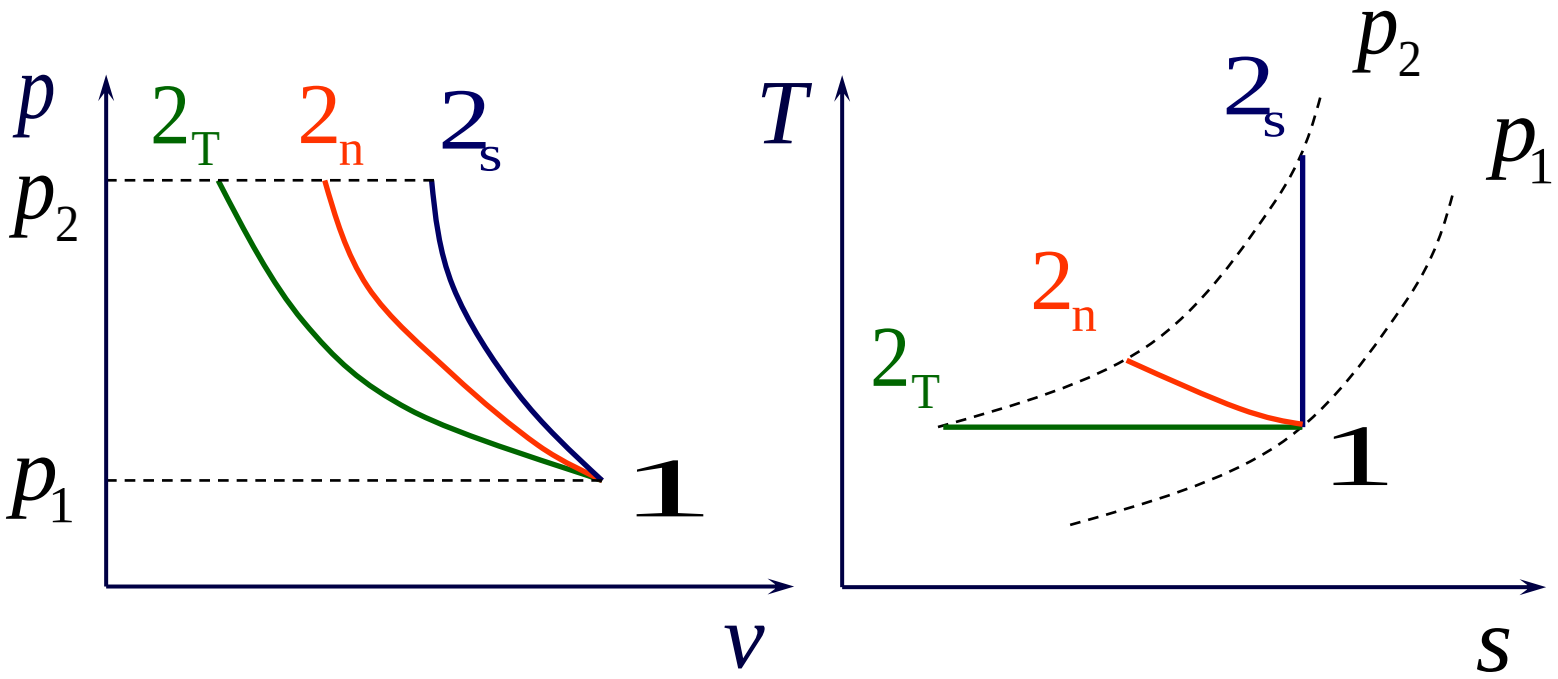
→ 技术功w_t

可能的压气过程

(1)特别快，来不及换热 (s) $n = k$

(2)特别慢，热全部散掉 (T) $n = 1$

(3)实际压气过程是 (n) $1 < n < k$

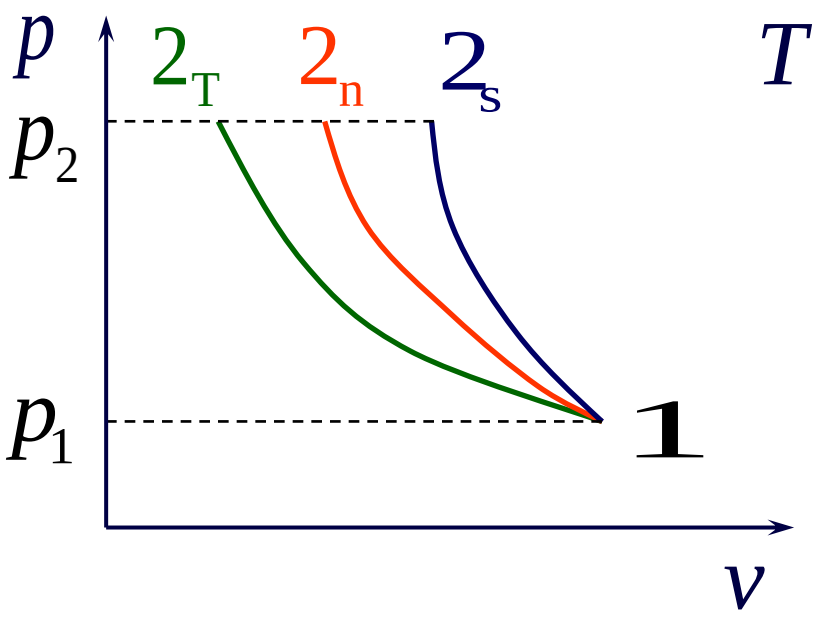


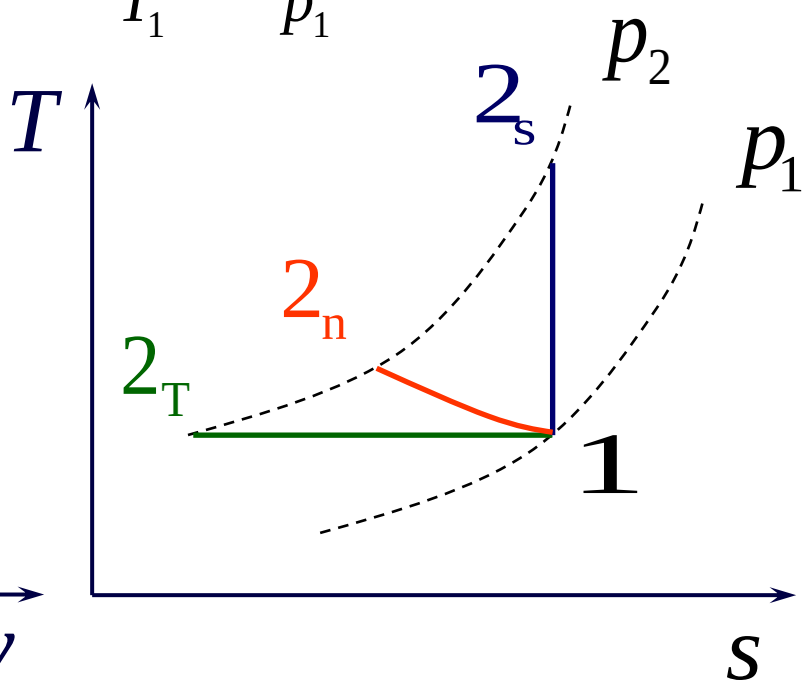
压缩过程中气体的状态变化

s $n = k$ $\quad \frac{p_2}{p_1} = (\frac{v_1}{v_2})^k \quad \frac{T_2}{T_1} = (\frac{p_2}{p_1})^{\frac{k-1}{k}}$

T $n = 1$ $\quad \frac{p_2}{p_1} = \frac{v_1}{v_2} \quad T_1 = T_2 = T$

n $1 < n < k$ $\quad \frac{p_2}{p_1} = (\frac{v_1}{v_2})^n \quad \frac{T_2}{T_1} = (\frac{p_2}{p_1})^{\frac{n-1}{n}}$



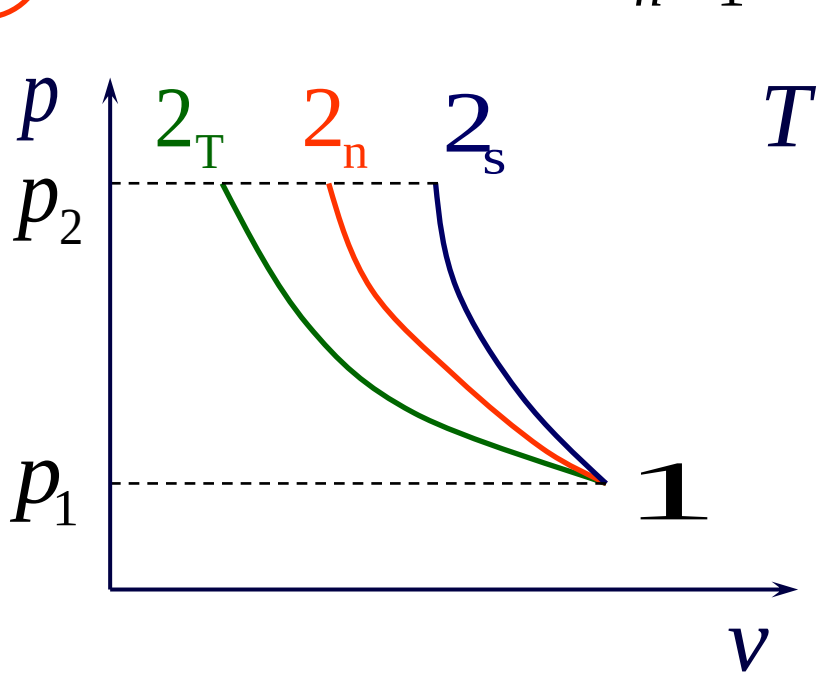


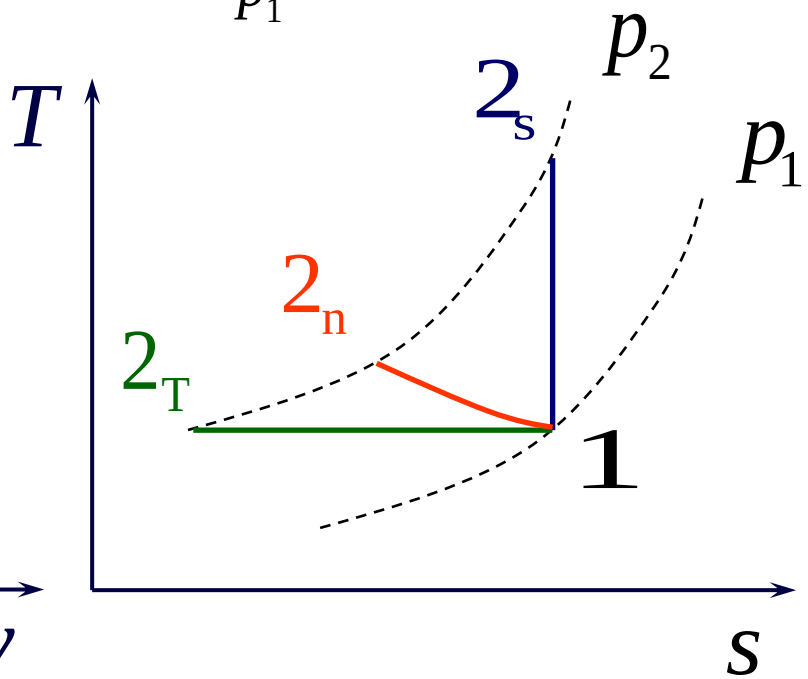
压缩功

s $n = k$ $\quad w_t = \frac{k}{k-1} p_1 v_1 [1 - (\frac{p_2}{p_1})^{\frac{k-1}{k}}] < 0$

T $n = 1$ $\quad w_t = RTIn \frac{p_1}{p_2} < 0$

n $1 < n < k$ $\quad w_t = \frac{n}{n-1} p_1 v_1 [1 - (\frac{p_2}{p_1})^{\frac{n-1}{n}}]$



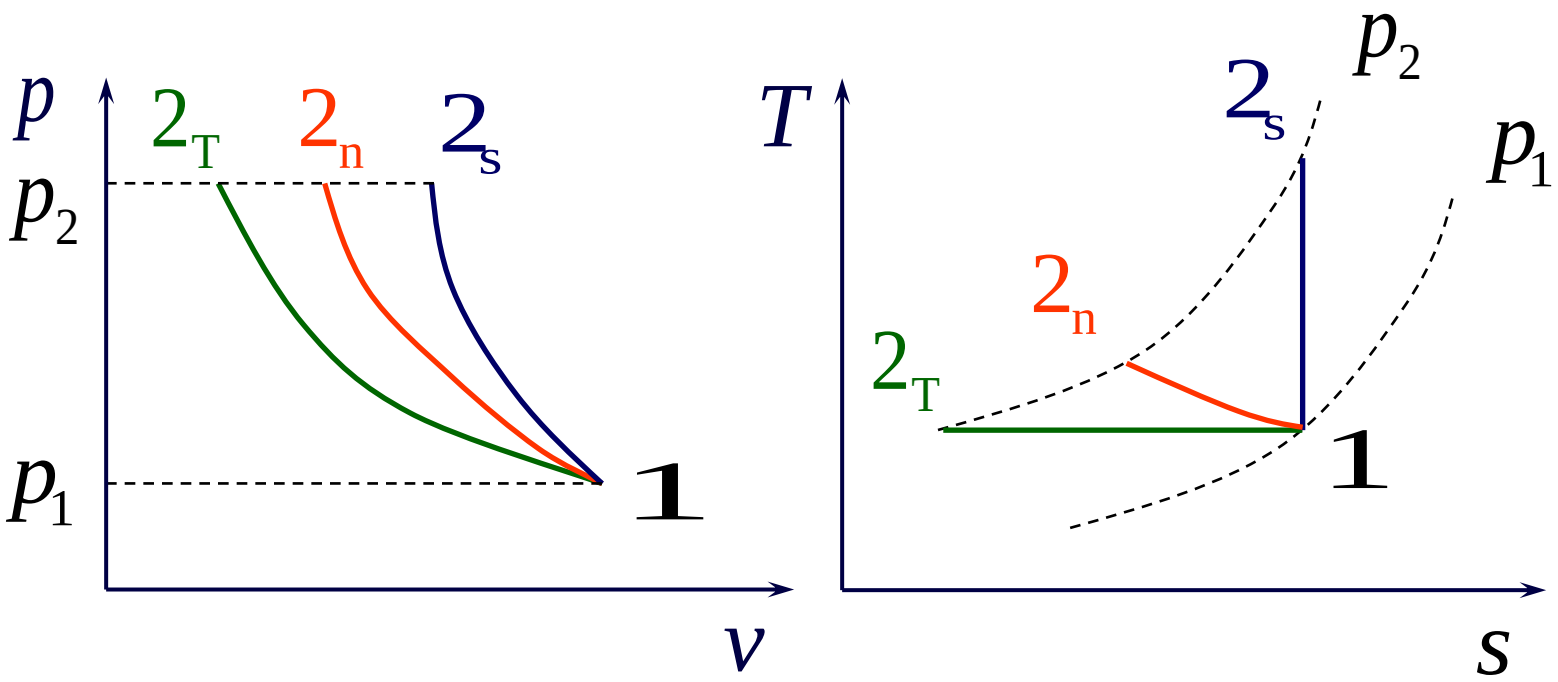


换热量

s $n = k$ $\quad q = 0$

T $n = 1$ $\quad q = RTIn\dfrac{p_1}{p_2} < 0$

n $1 < n < k$ $\quad q = c_n(T_2 - T_1) = \dfrac{n-k}{n-1}c_v T_1[(\dfrac{p_2}{p_1})^{\frac{n-1}{n}} - 1]$



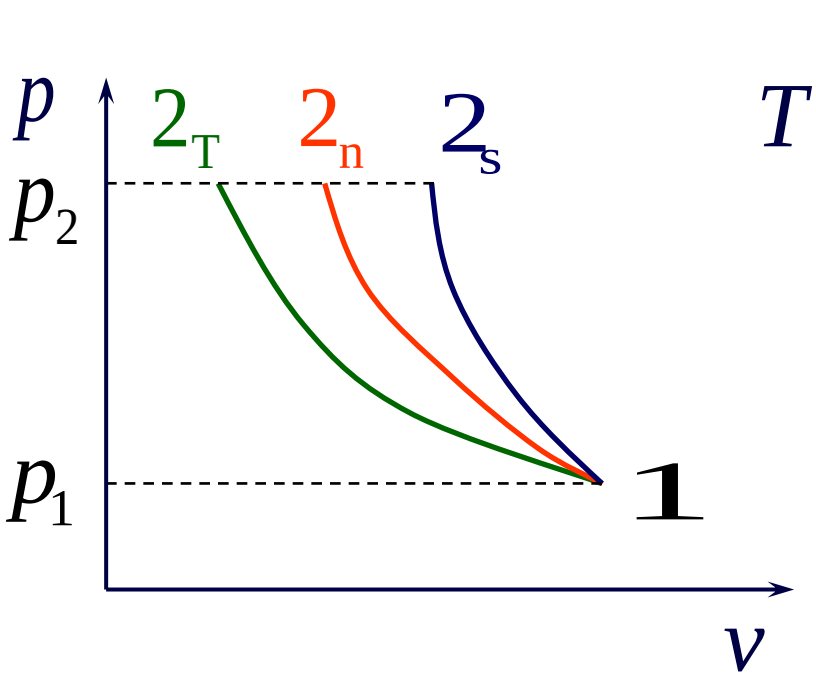
三种压气过程的参数对比

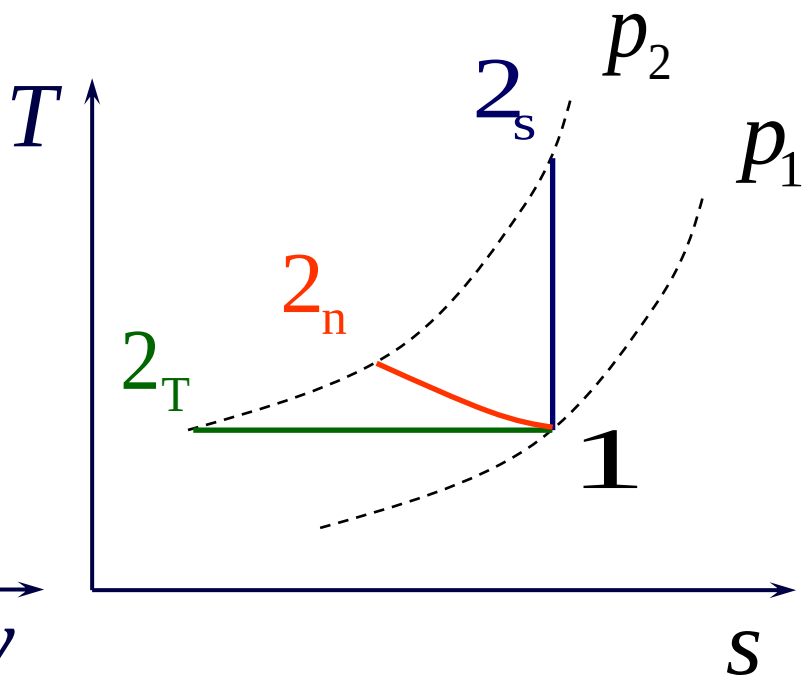
$$|w_{tT}| < |w_{tn}| < |w_{ts}|$$

$$T_1 = T_{2T} < T_{2n} < T_{2s}$$

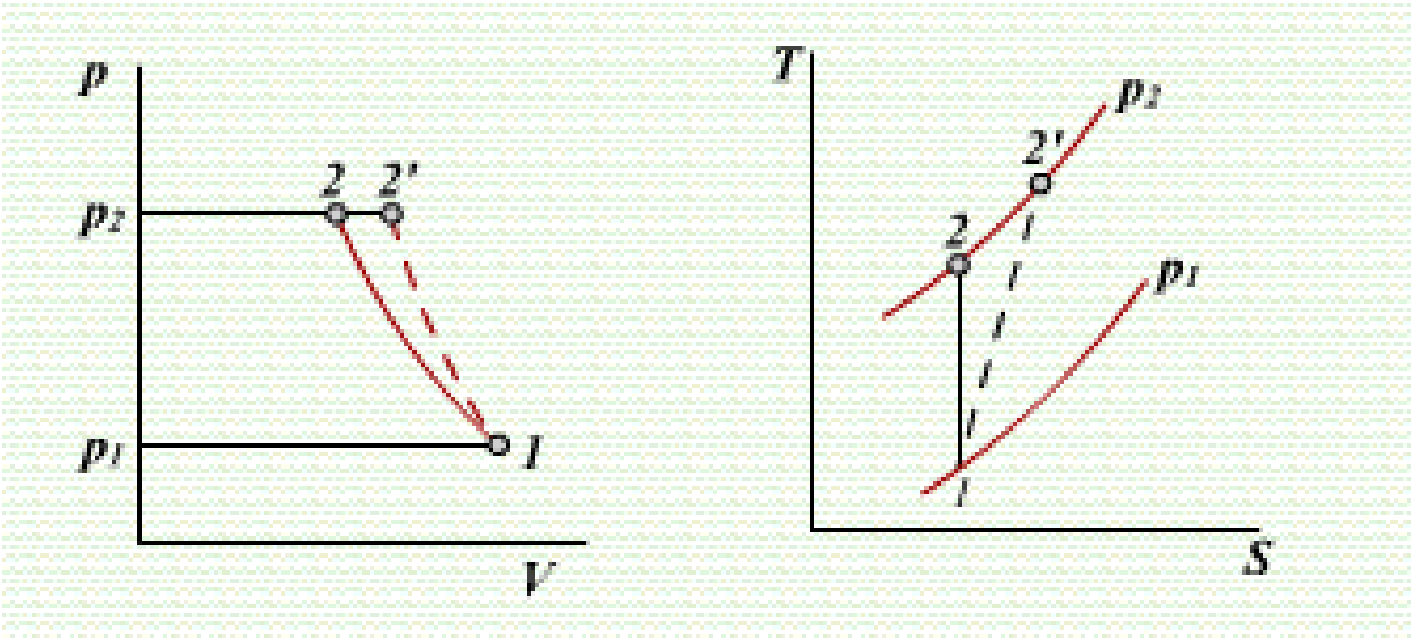
$$v_{2T} < v_{2n} < v_{2s}$$

$$|q_T| > |q_n| > |q_s| = 0$$





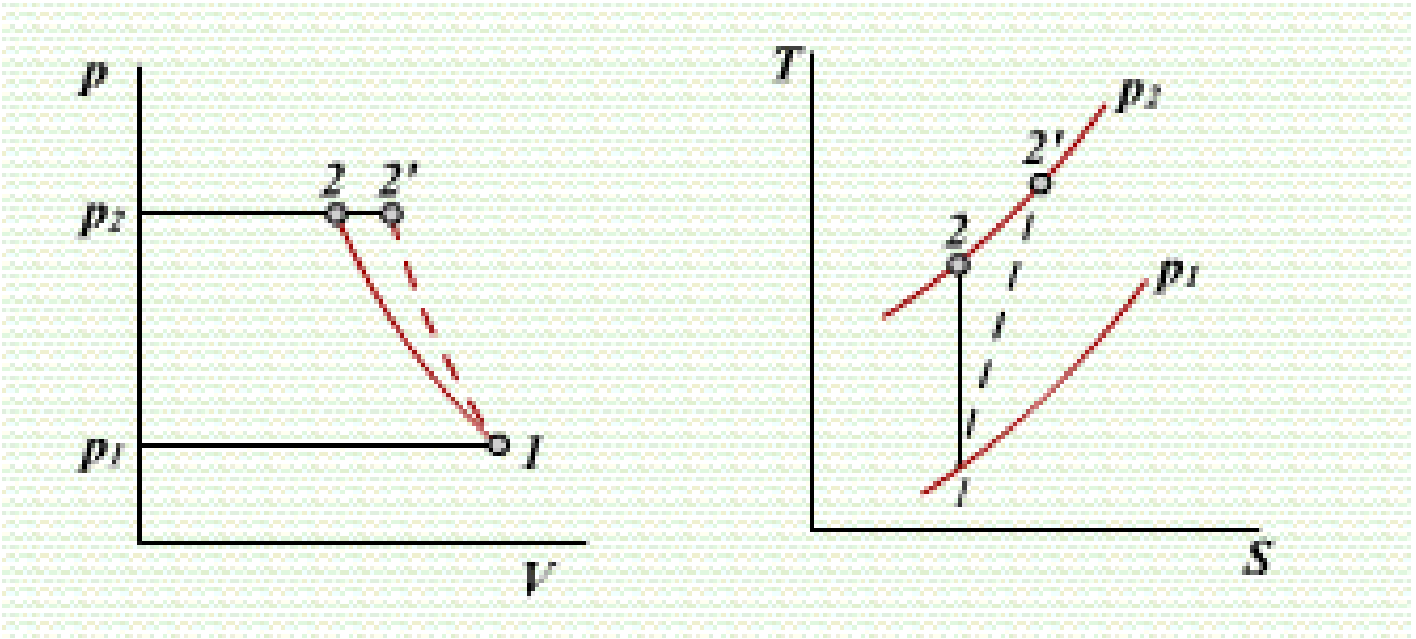
摩擦对压缩过程的影响



(1) 初终态状态参数间的关系

$$\frac{p_2}{p_1}=\left(\frac{v_1}{v_{2'}}\right)^n \qquad \frac{T_{2'}}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} \qquad k<n<\infty$$

摩擦对压缩过程的影响

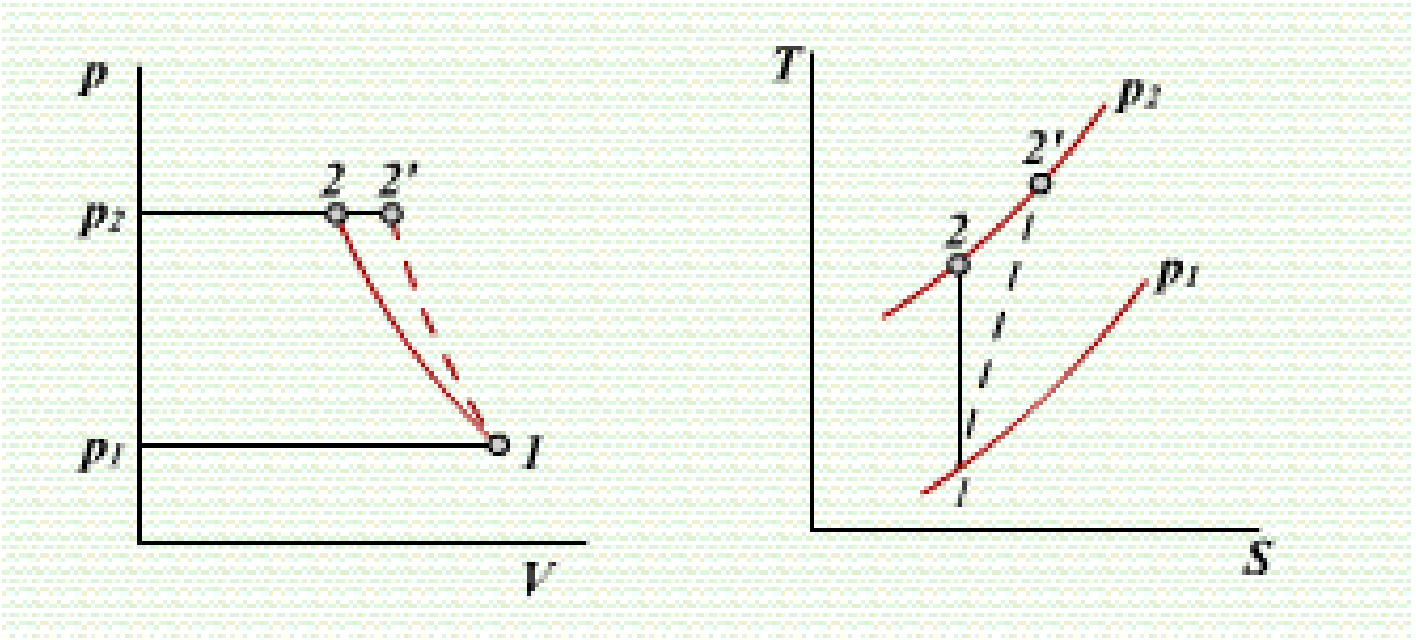


(2) 压缩功

$$w_t' = -\int v dp = \frac{n}{n-1}\left(p_1 v_1 - p_{2'} v_{2'}\right) = \frac{n}{n-1} R\left(T_1 - T_{2'}\right)$$

$$= \frac{n}{n-1} R T_1 \left[1 - \frac{T_{2'}}{T_1}\right] = \frac{n}{n-1} R T_1 \left[1 - \left(\frac{p_{2'}}{p_1}\right)^{\frac{n-1}{n}}\right]$$

摩擦对压缩过程的影响

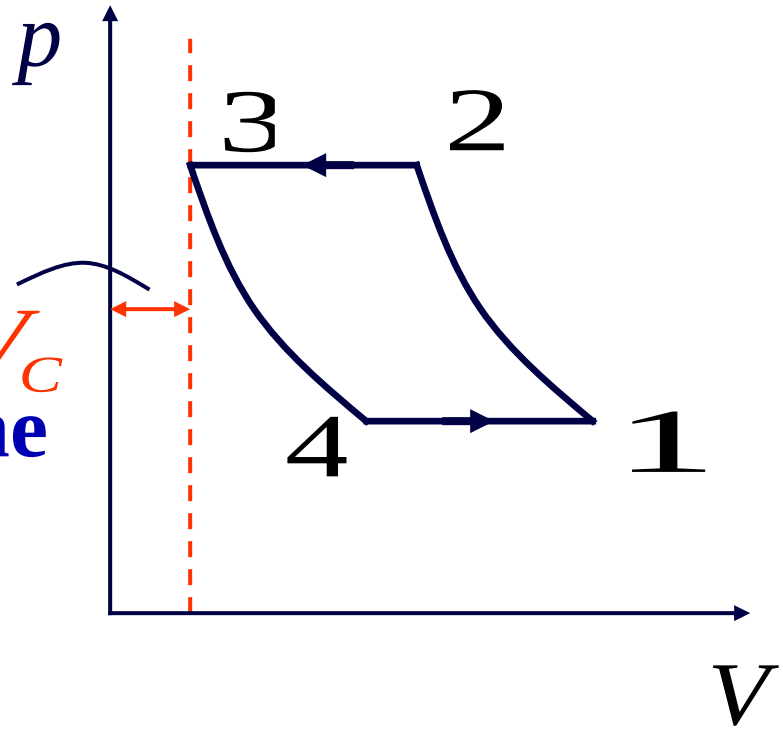


压气机绝热效率 $\eta_{c,s} = \dfrac{w_t}{w_t'} = \dfrac{h_1 - h_2}{h_1 - h_{2'}} < 1$

定比热理想气体 $\eta_{c,s} = \dfrac{c_p(T_1 - T_2)}{c_p(T_1 - T_{2'})} = \dfrac{T_1 - T_2}{T_1 - T_{2'}}$

§10-3 活塞式压气机余隙容积的影响

避免活塞与进排气阀碰撞，留有空隙



余隙容积 V_C

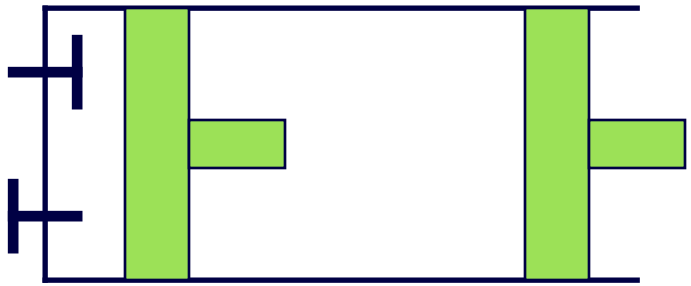
Clearance volume

1⇨2 压缩过程

2⇨3 排气，状态未变

3⇨4 残留气体膨胀

4⇨1 进新气，状态未变



§10-3 活塞式压气机余隙容积的影响

$V_h = V_1 - V_3$ 活塞排量

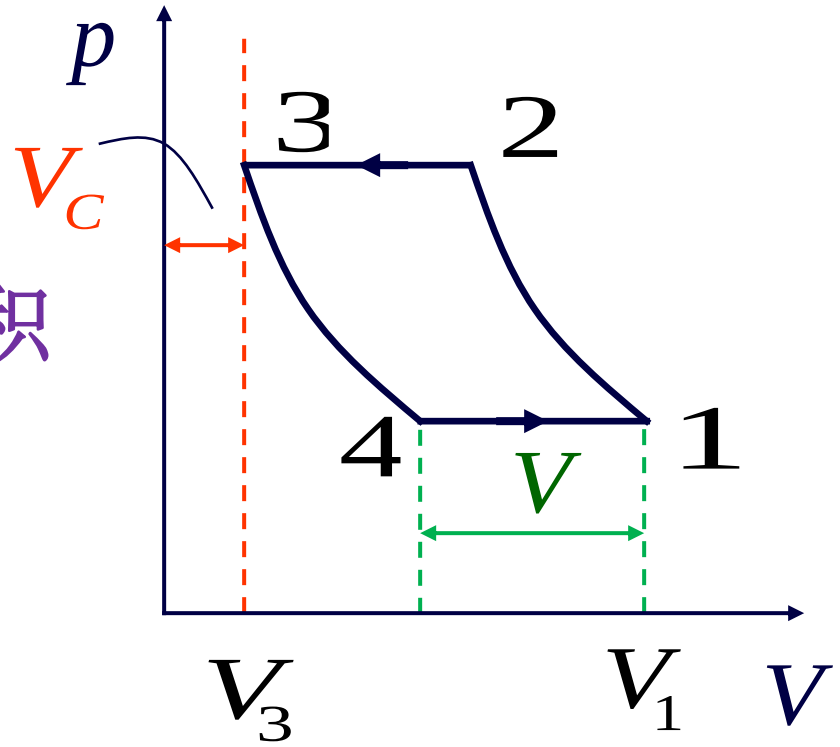
排量容积

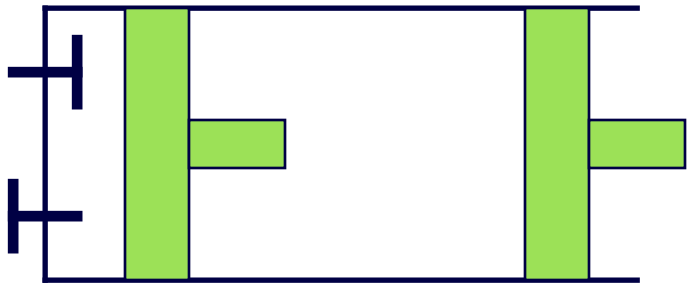
$V = V_1 - V_4$ 有效进气容积

$p_1 V = mRT_1$

新气量

产气量





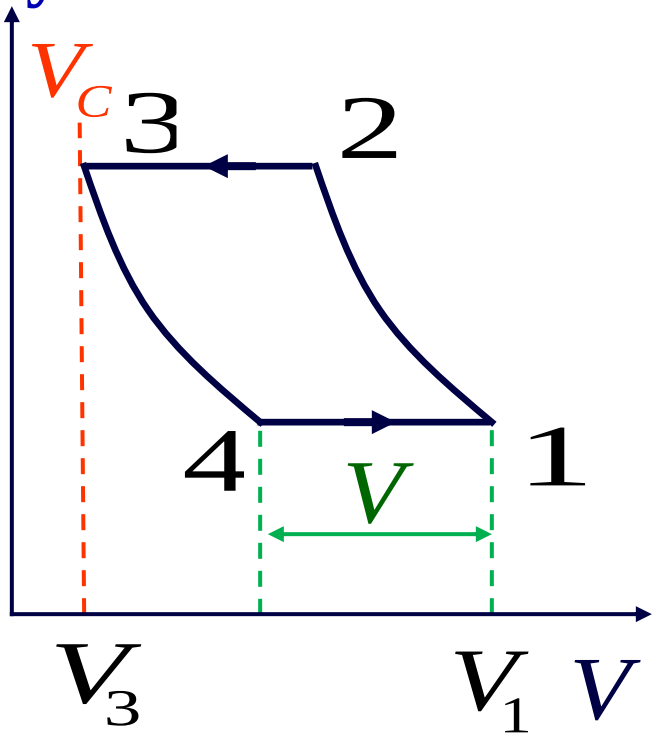
研究V_C对产气量和耗功的影响

余隙容积V_C对产气量的影响

定义容积效率 **Volumetric efficiency**

$$\eta_V = \frac{V}{V_h} = \frac{V_1 - V_4 + V_3 - V_3}{V_1 - V_3}$$

$$= 1 - \frac{V_4 - V_3}{V_1 - V_3} = 1 - \frac{V_3}{V_1 - V_3}\left(\frac{V_4}{V_3} - 1\right)$$



令 $c = \frac{V_C}{V_h}$ 余隙比

工程上一般＝**0.03~0.08**

$$\eta_V = 1 - c\left[\left(\frac{p_3}{p_4}\right)^{\frac{1}{n}} - 1\right] = 1 - c\left[\left(\frac{p_2}{p_1}\right)^{\frac{1}{n}} - 1\right]$$

余隙容积V_C对产气量的影响

$$\eta_V = \frac{V}{V_h} = 1 - c\left[\left(\frac{p_2}{p_1}\right)^{\frac{1}{n}} - 1\right]$$

$$c = \frac{V_C}{V_h}$$

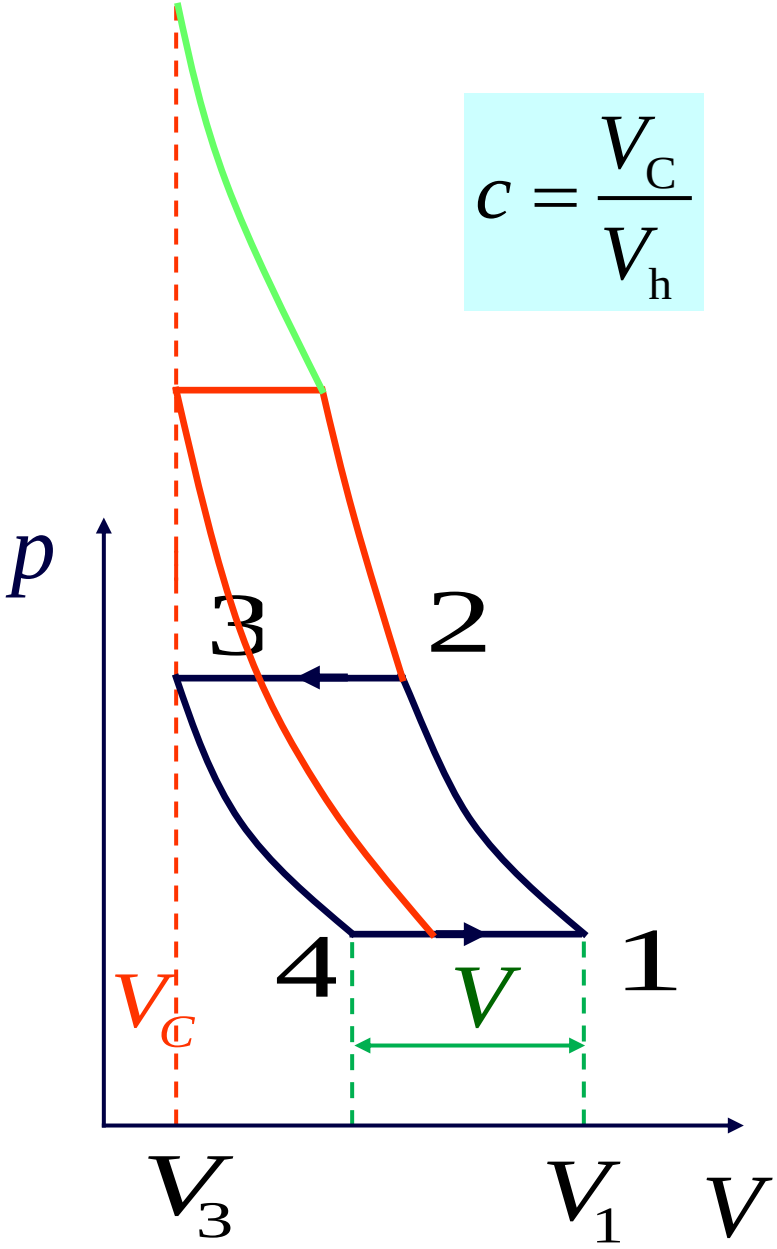
讨论：

(1) $\frac{p_2}{p_1}$ 一定，c ↑ η_V ↓

(2) c 和 n 一定，$\frac{p_2}{p_1}$ ↑ η_V ↓

极限 $\eta_V = 0$

故 单级增压比不超过8-9

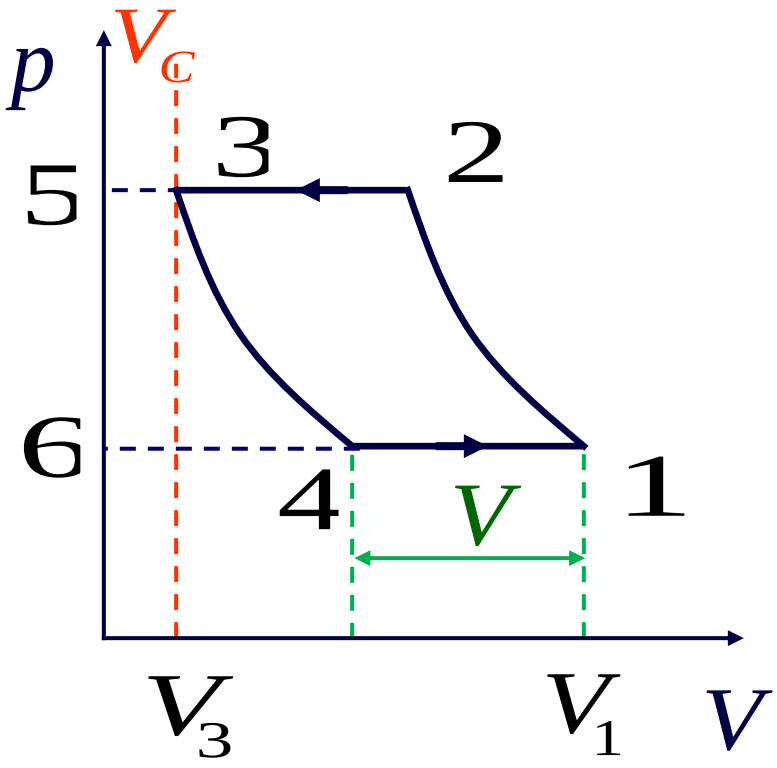


余隙容积V_C对理论压气功的影响

|功|＝面积12341

＝面积12561－面积43564

设1→2和3→4两过程n相同



$$W_t = \frac{n}{n-1} p_1 V_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right]$$

$$-\frac{n}{n-1} p_1 V_4 \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right]$$

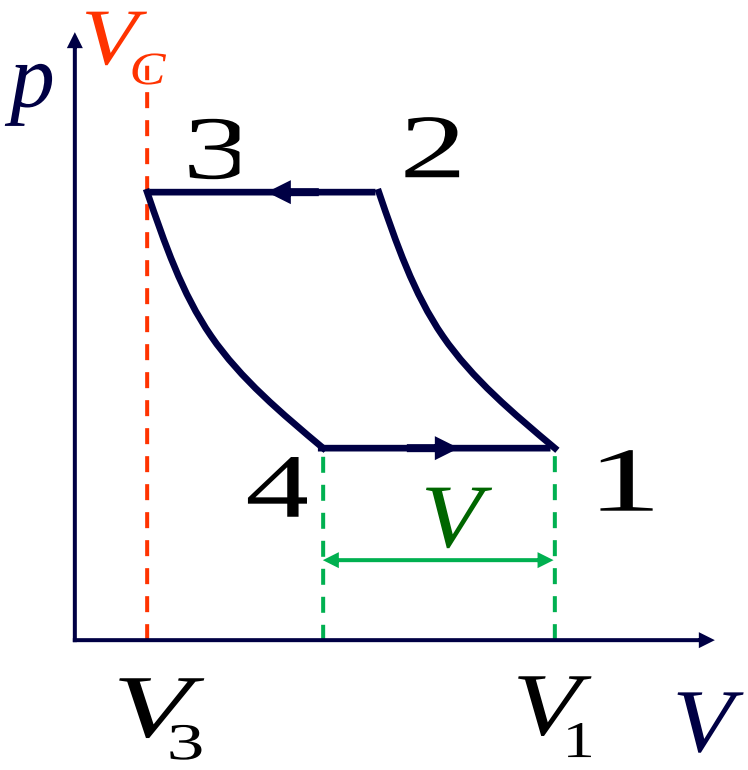
$$p_2 = p_3$$

$$p_1 = p_4$$

余隙容积V_C对理论压气功的影响

$$W_t = \frac{n}{n-1} p_1 (V_1 - V_4) \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right]$$

$$= \frac{n}{n-1} p_1 V \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right]$$



由 $p_1 V = m R T_1$ 得：

$$w_t = \frac{W_t}{m} = \frac{n}{n-1} R T_1 \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right]$$

余隙对单位产气量耗功不影响

余隙影响例题

已知：$V_h = 0.4455m^3, V_3 = 0.0223m^3$

$p_1 = 1\text{bar}, t_1 = 15^oC, p_2 = 3.5\text{bar}, n = 1.2$

求（1）有余隙时的产气量和耗功

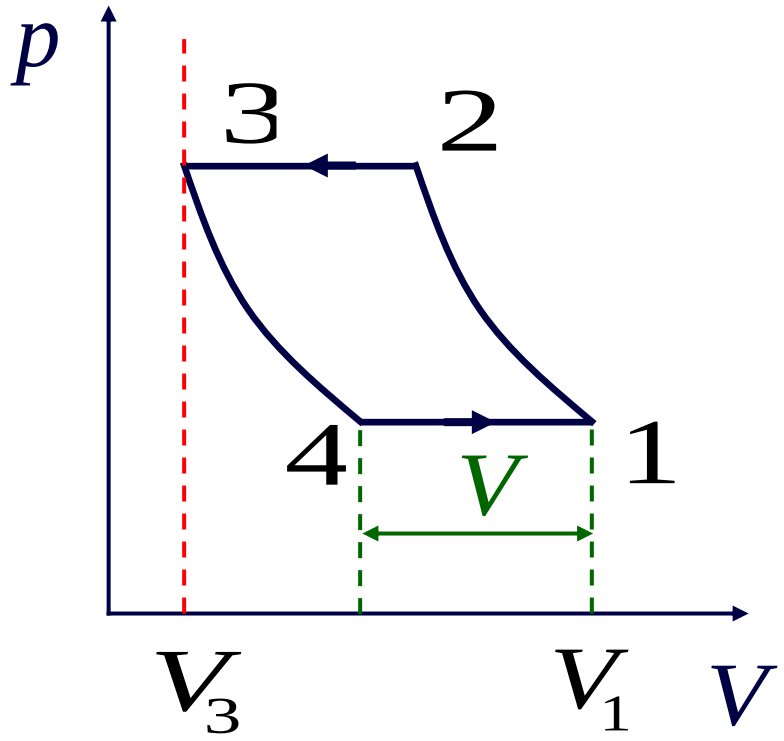
（2）无余隙时的产气量和耗功

解：（1）有效容积

$$V = V_1 - V_4 = V_h + V_3 - V_4$$

$$= V_h + V_3 - V_3\left(\frac{p_2}{p_1}\right)^{\frac{1}{n}}$$

$$= 0.4045m^3$$



余隙影响例题（续）

$$V = 0.4045\ \mathrm{m}^3$$

$$W_t = \frac{n}{n-1} p_1 V \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right] = -56.35\ \mathrm{kJ}$$

$$m = \frac{p_1 V}{R T_1} = 0.489\ \mathrm{kg}$$

$$w_t = \frac{W_t}{m} = -115\ \mathrm{kJ/kg}$$

余隙影响例题（续）

（2）无余隙时的产气量和耗功

有效容积 $V = V_1 = V_h + V_3 = 0.4678 m^3$

$$W_t' = \frac{n}{n-1} p_1 V \left[1 - \left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} \right] = -65.17 kJ$$

$$m' = \frac{p_1 V}{R T_1} = 0.566 kg \qquad m = 0.489 kg$$

$$w_t' = \frac{W_t}{m} = -115 \text{ kJ/kg} \qquad w_t = -115 \text{ kJ/kg}$$

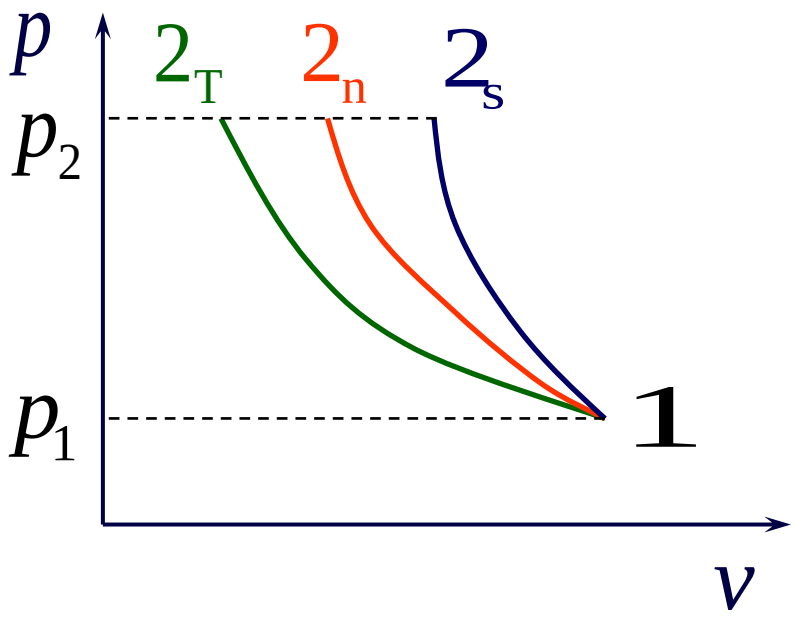
§10-4 多极压缩及中间冷却

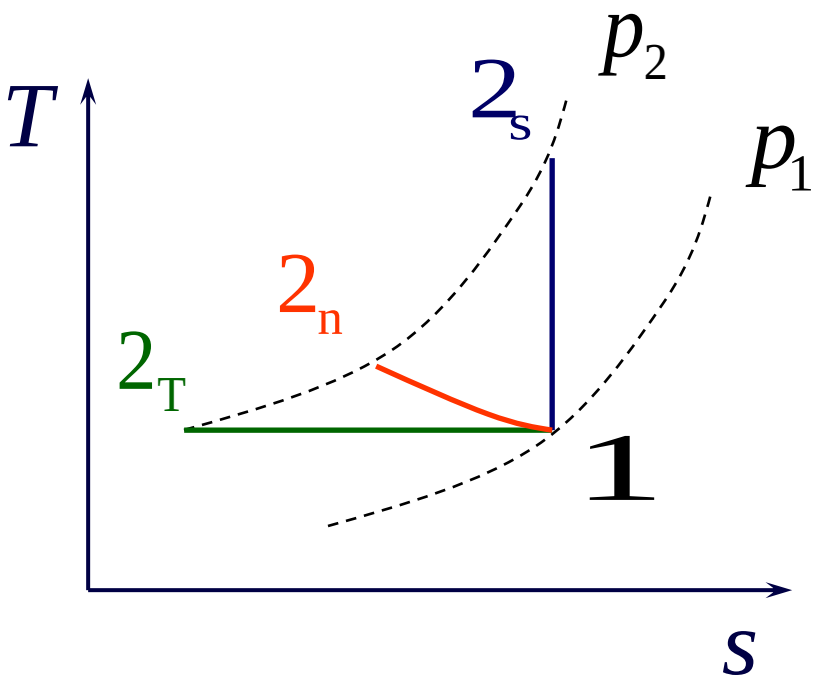
$$|w_{tT}| < |w_{tn}| < |w_{ts}|$$

$$T_1 = T_{2T} < T_{2n} < T_{2s}$$

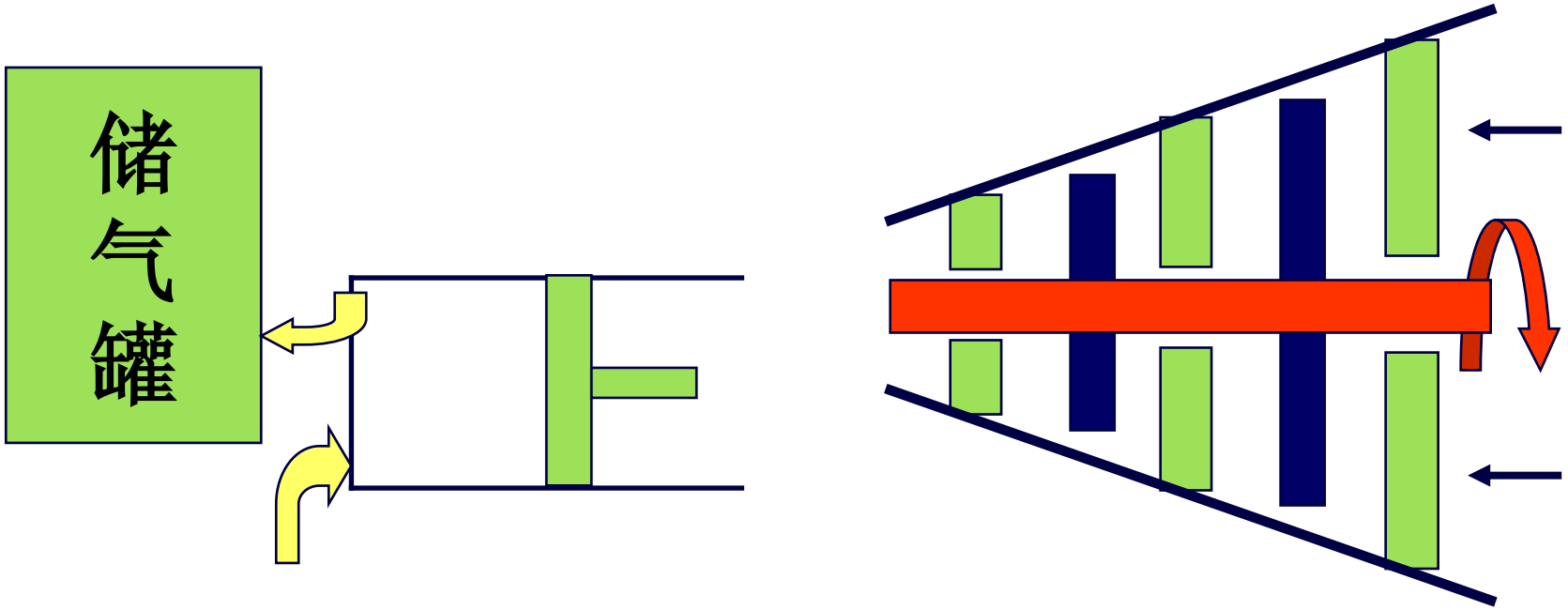
等温过程最小

要冷却





压气机的结构与冷却

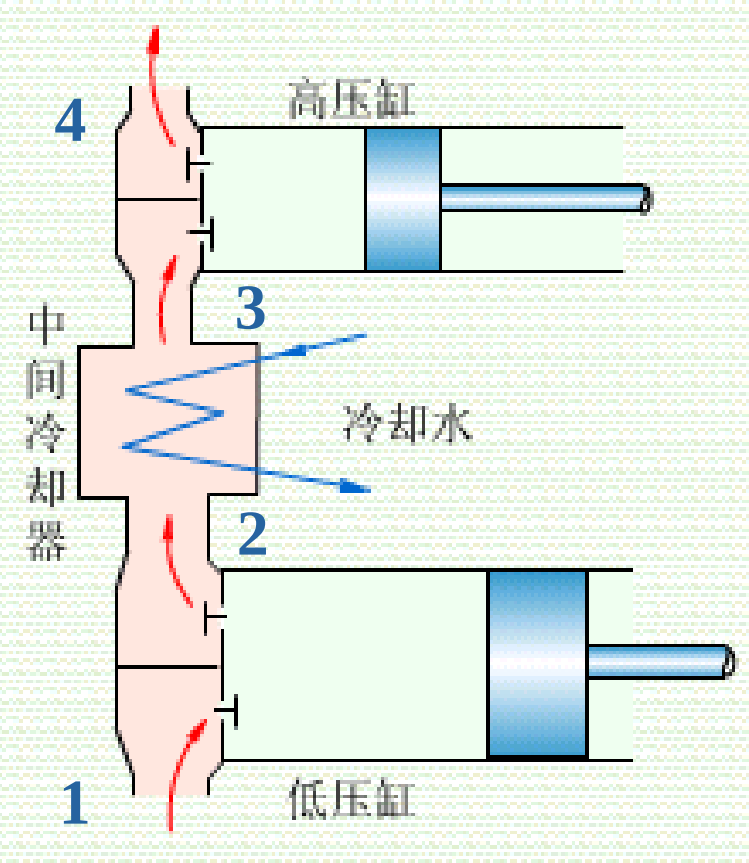


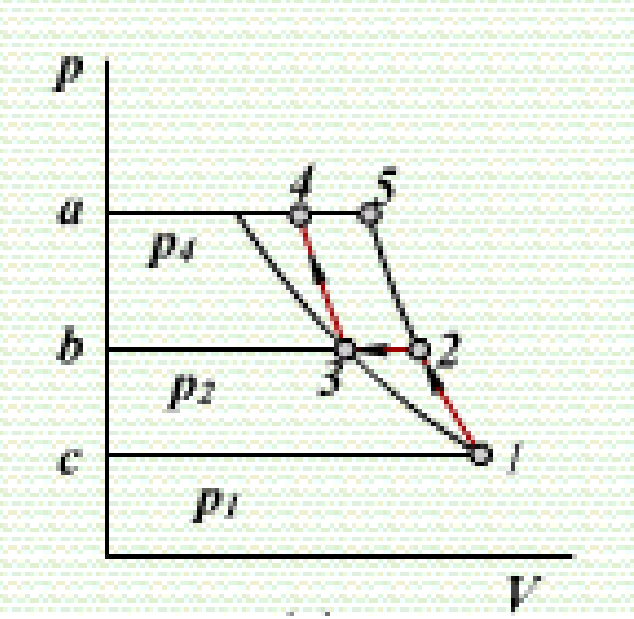
活塞式压气机易于实现放热压缩

多级压缩中间冷却

Multistage compression with intercooling

以两级压缩为例



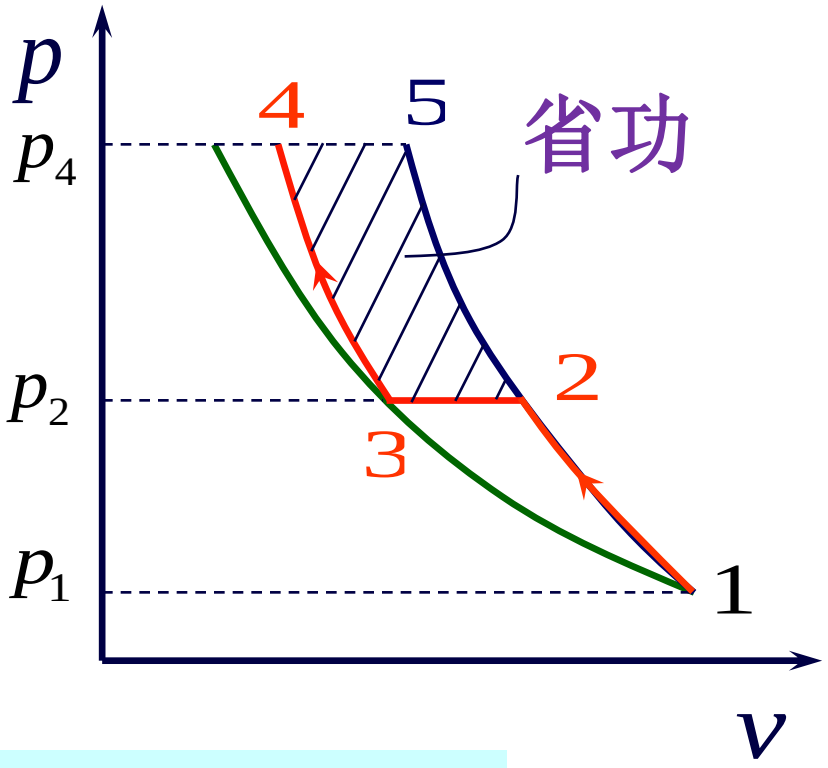


压缩功

$$w_{分级} = w^{I}_{t(n)} + w^{II}_{t(n)}$$

$$= \frac{n}{n-1} RT_1 [1 - (\frac{p_2}{p_1})^{\frac{n-1}{n}}]$$

$$+ \frac{n}{n-1} RT_3 [1 - (\frac{p_4}{p_3})^{\frac{n-1}{n}}]$$



$$w_{分级} = \frac{n}{n-1} RT_1 [2 - (\frac{p_2}{p_1})^{\frac{n-1}{n}} - (\frac{p_4}{p_2})^{\frac{n-1}{n}}]$$

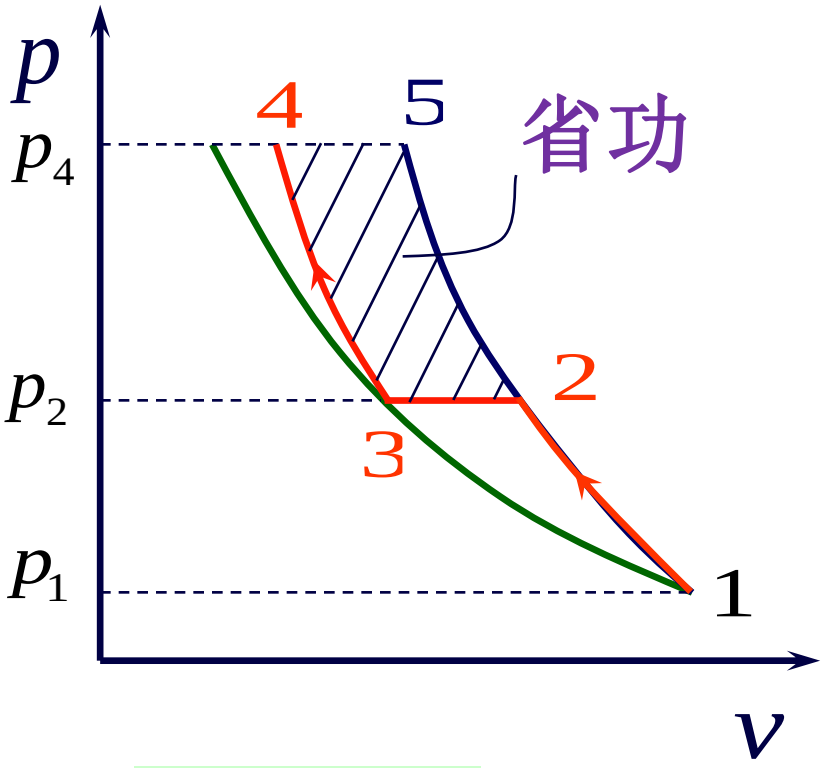
最佳增压比

压缩功最小 $\dfrac{\partial w_{分级}}{\partial p_2}=0$

$$p_2=\sqrt{p_1 p_4} \qquad \frac{p_2}{p_1}=\frac{p_4}{p_2}$$

$$\pi=\frac{p_2}{p_1}=\frac{\sqrt{p_1 p_4}}{p_1}$$

$$=\sqrt{\frac{p_4}{p_1}}=\sqrt{\frac{p_{终}}{p_{初}}}$$

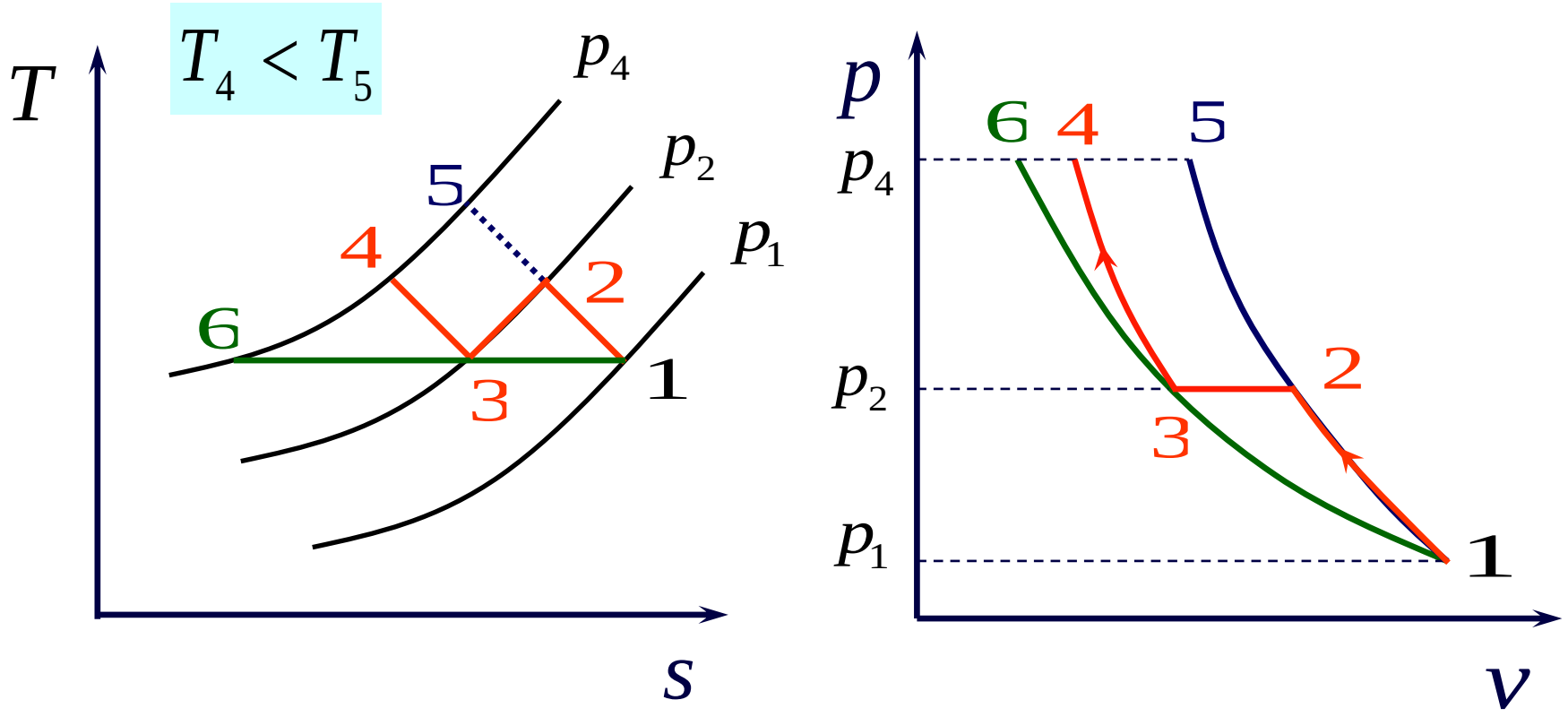


可证明

若m级

$$\pi=\sqrt[m]{\frac{p_{终}}{p_{初}}}$$

分级压缩的其它好处



润滑油要求t <160~180°C，高压压气机必须分级

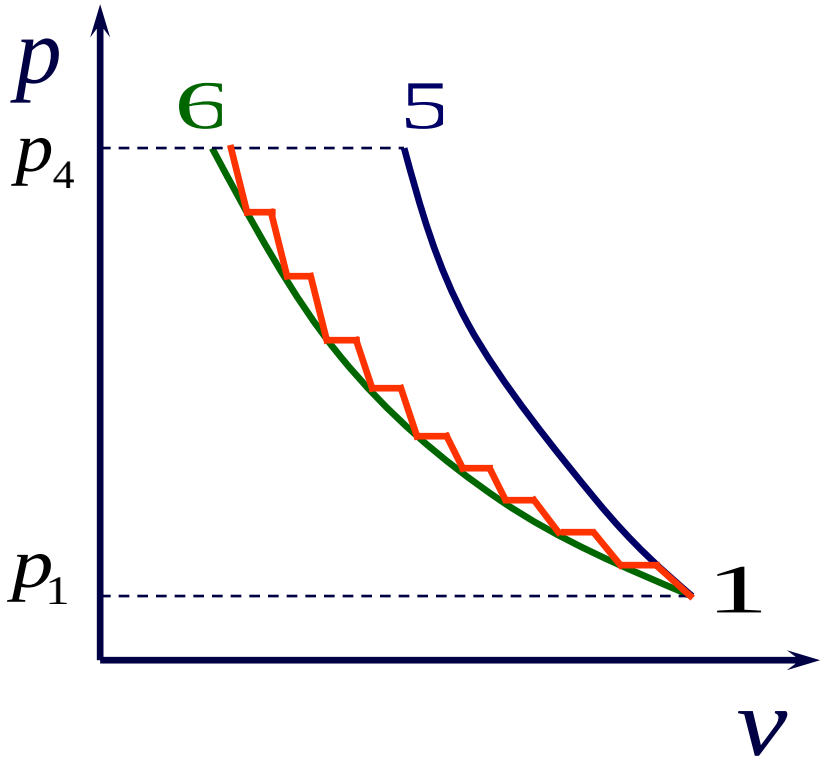
另外，多级压缩的容积效率由于各级增压比的减少而比单级压缩时高

分级压缩的级数

分级 { 省功
降低出口温度

多级压缩达到
无穷多级 → T

{ (1) 不可能实现
(2) 结构复杂(成本高)



所以，一般采用 2 ~ 4 级压缩

第十章 完

End of Chapter Ten

第十章 小结 Summary

1. 压气机的作用、分类、工作原理及热力过程分析

2. 活塞式压气机余隙容积的影响

3. 多级压缩及中间冷却

